```
                                    MM        MM  FFFFFFFFFFF       11
                                   MMM      MMM  FFFFFFFFFFF      111
                                  MMMM    MMMM  FF               1111
                                 MM MM  MM MM  FF                 11
                                MM  MMMM  MM  FF                  11
                               MM   MM   MM  FFFFFFF              11
                              MM        MM  FFFFFFF              11
                             MM        MM  FF                   11
                            MM        MM  FF                   11
                           MM        MM  FF             1111111111
                          MM        MM  FF              1111111111


           SSSSSSSSS    6666666666        444                                  AAAAAAAAAA
          SSSSSSSSSSS  666666666666      4444                                 AAAAAAAAAAAA
          SS        SS  66        66     44 44                                AA        AA
          SS            66              44  44                                AA        AA
          SSS           66             44   44                                AA        AA
           SSSSSSSSS    66666666666   44444444444                             AAAAAAAAAAAA
            SSSSSSSSS   666666666666  444444444444                            AAAAAAAAAAAA
                   SSS  66        66        44                                 AA        AA
                    SS  66        66        44                                 AA        AA
          SS        SS  66        66        44                                 AA        AA
          SSSSSSSSSSS  666666666666        44                                 AA        AA
           SSSSSSSSS    6666666666        44                                 AA        AA


             PPPPPPPPPP   RRRRRRRRRR    IIIIIIIII   MM        MM  FFFFFFFFFF  OOOOOOOOO  RRRRRRRRR   HH        HH
            PPPPPPPPPPP  RRRRRRRRRRR   IIIIIIIII   MMM      MMM  FFFFFFFFFF  OOOOOOOOO  RRRRRRRRRR  HH        HH
           PP        PP  RR        RR     II       MMMM    MMMM  FF          OO       OO  RR       RR  HH        HH
          PP        PP  RR        RR      II       MM MM  MM MM  FF          OO       OO  RR       RR  HH        HH
         PP        PP  RR        RR       II       MM  MMMM  MM  FF          OO       OO  RR       RR  HH        HH
        PPPPPPPPPPP  RRRRRRRRRRR          II       MM   MM   MM  FFFFFFF     OO       OO  RRRRRRRRRRR  HHHHHHHHHHHH
       PPPPPPPPPP   RRRRRRRRRR            II       MM        MM  FFFFFFF     OO       OO  RRRRRRRRRR   HHHHHHHHHHHH
      PP           RR    RR              II       MM        MM  FF          OO       OO  RR    RR     HH        HH
     PP           RR     RR             II       MM        MM  FF          OO       OO  RR     RR    HH        HH
    PP           RR      RR         IIIIIIIII   MM        MM  FF          OOOOOOOOOOO  RR      RR   HH        HH
   PP           RR       RR        IIIIIIIII   MM        MM  FF          OOOOOOOOOOO  RR       RR  HH        HH


                  JJJJJJJJJ      11       3333333333                            AAAAAAAAAA
                  JJJJJJJJJ     111      333333333333                          AAAAAAAAAAAA
                      JJ       1111      33       33                           AA        AA
                      JJ         11              33                            AA        AA
                      JJ         11              33                            AA        AA
                      JJ         11           3333                             AAAAAAAAAAAA
                      JJ         11           3333                             AAAAAAAAAAAA
                      JJ         11              33                            AA        AA
                JJ    JJ         11              33                            AA        AA
                JJ    JJ         11      33      33                            AA        AA
                JJJJJJJ      1111111111  333333333333                          AA        AA
                 JJJJJ       1111111111   3333333333                           AA        AA

****A  START  JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13  START  A****
****A  START  JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13  START  A****
****A  START  JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13  START  A****
****A  START  JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13  START  A****
```

```
21.17.20 JOB   13  $HASP373 PRIMFORH STARTED - INIT  1 - CLASS A - SYS TK4-
21.17.20 JOB   13  IEF403I PRIMFORH - STARTED - TIME=21.17.20
21.17.20 JOB   13  IEFACTRT - Stepname  Procstep  Program   Retcode
21.17.20 JOB   13  PRIMFORH   PRIMES    FORT      IEKAA00   RC= 0000
21.17.21 JOB   13  PRIMFORH   PRIMES    GO        LOADER    RC= 0000
21.17.21 JOB   13  IEF404I PRIMFORH - ENDED - TIME=21.17.21
21.17.21 JOB   13  $HASP395 PRIMFORH ENDED


------ JES2 JOB STATISTICS ------


 26 NOV 21 JOB EXECUTION DATE


       204 CARDS READ


       339 SYSOUT PRINT RECORDS


         0 SYSOUT PUNCH RECORDS


      0.00 MINUTES EXECUTION TIME
```

```
1      //PRIMFORH JOB (FORTRAN),                                          JOB   13
       //             'Eratosthenes Sieve',
       //             CLASS=A,
       //             MSGCLASS=A,
       //             REGION=9000K,TIME=1440,
       //             MSGLEVEL=(1,1),
       //           USER=HERC01,PASSWORD=            GENERATED BY GDL
       ***************************************************************
       ***
       *** Name: SYS2.JCLLIB(PRIMFORH)
       ***
       *** Desc: Sieve of Eratosthenes programmed in FORTRAN,
       ***       compiled using the IBM OS/360 FORTRAN H Level 21.8 compiler.
       ***       All prime numbers up to the value entered via
       ***       //GO.SYSIN DD are computed.
       ***
       ***************************************************************
2      //PRIMES    EXEC FORTHCLD,PARM.GO='SIZE=9000000'
3      XXFORTHCLD PROC SOUT='*'                                          00000100
4      XXFORT     EXEC PGM=IEKAA00,PARM=(SOURCE,MAP)                     00000200
5      XXSYSPRINT DD SYSOUT=&SOUT                                        00000300
6      XXSYSPUNCH DD SYSOUT=B                                            00000400
7      XXSYSLIN   DD DSNAME=&LOADSET,UNIT=SYSDA,DISP=(MOD,PASS),         00000500
       XX            SPACE=(400,(200,50),RLSE)                          00000600
8      //FORT.SYSIN DD *
9      XXGO       EXEC PGM=LOADER,PARM=(MAP),COND=(4,LT,FORT)            00000700
10     XXSYSLIB   DD DSNAME=SYS1.FORTLIB,DISP=SHR                        00000800
11     XXSYSLOUT  DD SYSOUT=&SOUT                                        00000900
12     XXSYSLIN   DD DSNAME=*.FORT.SYSLIN,DISP=(OLD,PASS)                00001000
13     XXFT05F001 DD DDNAME=SYSIN                                        00001100
14     //GO.FT06F001 DD SYSOUT=*,DCB=(RECFM=FBA,LRECL=166,BLKSIZE=16600)
       X/FT06F001 DD SYSOUT=&SOUT                                        00001200
15     XXFT07F001 DD SYSOUT=B                                            00001300
16     //GO.SYSIN DD *
```

```
 STMT NO. MESSAGE
-
    5      IEF653I SUBSTITUTION JCL - SYSOUT=*
   11      IEF653I SUBSTITUTION JCL - SYSOUT=*
   14      IEF653I SUBSTITUTION JCL - SYSOUT=*
IEF236I ALLOC. FOR PRIMFORH FORT PRIMES
IEF237I JES2 ALLOCATED TO SYSPRINT
IEF237I JES2 ALLOCATED TO SYSPUNCH
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I JES2 ALLOCATED TO SYSIN
IEF142I PRIMFORH FORT PRIMES - STEP WAS EXECUTED - COND CODE 0000
IEF285I   JES2.JOB00013.SO0103                    SYSOUT
IEF285I   JES2.JOB00013.SO0104                    SYSOUT
IEF285I   SYS21330.T211720.RA000.PRIMFORH.LOADSET      PASSED        *-------39
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00013.SI0101                    SYSIN
IEF373I STEP /FORT    / START 21330.2117
IEF374I STEP /FORT    / STOP  21330.2117 CPU    0MIN 00.05SEC SRB    0MIN 00.01SEC VIRT  9064K SYS   232K

*------------------------------------------------------------------------------------------------------------------------------*
*     1. Jobstep of job: PRIMFORH    Stepname: FORT        Program name: IEKAA00    Executed on 26.11.21 from 21.17.20 to 21.17.20 *
*         elapsed time  00:00:00,12                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,06              Virtual Storage used:   9064K        Page-out:      0                          *
*           corr. CPU:  00:00:00,06   CPU time has been corrected by  1 / 1,0  multiplier                                      *
*                                                                                                                              *
*     I/O Operation                                                                                                            *
*     Number of records read via DD * or DD DATA:    180                                                                       *
*     DMY.......0 DMY.......0 180......39 DMY.......0                                                                          *
*                                                                                                                              *
*                                       Charge for step (w/o SYSOUT):          0,10                                        *
*------------------------------------------------------------------------------------------------------------------------------*
IEF236I ALLOC. FOR PRIMFORH GO PRIMES
IEF237I 148  ALLOCATED TO SYSLIB
IEF237I JES2 ALLOCATED TO SYSLOUT
IEF237I 180  ALLOCATED TO SYSLIN
IEF237I JES2 ALLOCATED TO FT05F001
IEF237I JES2 ALLOCATED TO FT06F001
IEF237I JES2 ALLOCATED TO FT07F001
IEF142I PRIMFORH GO PRIMES - STEP WAS EXECUTED - COND CODE 0000
IEF285I   SYS1.FORTLIB                            KEPT          *-------65
IEF285I   VOL SER NOS= MVSRES.
IEF285I   JES2.JOB00013.SO0105                    SYSOUT
IEF285I   SYS21330.T211720.RA000.PRIMFORH.LOADSET      PASSED        *-------40
IEF285I   VOL SER NOS= WORK02.
IEF285I   JES2.JOB00013.SI0102                    SYSIN
IEF285I   JES2.JOB00013.SO0106                    SYSOUT
IEF285I   JES2.JOB00013.SO0107                    SYSOUT
IEF373I STEP /GO      / START 21330.2117
IEF374I STEP /GO      / STOP  21330.2117 CPU    0MIN 00.04SEC SRB    0MIN 00.01SEC VIRT  8824K SYS   224K

*------------------------------------------------------------------------------------------------------------------------------*
*     2. Jobstep of job: PRIMFORH    Stepname: GO          Program name: LOADER     Executed on 26.11.21 from 21.17.20 to 21.17.21 *
*         elapsed time  00:00:00,10                    CPU-Identifier:  TK4-          Page-in:      0                          *
*             CPU time  00:00:00,05              Virtual Storage used:   8824K        Page-out:      0                          *
*           corr. CPU:  00:00:00,05   CPU time has been corrected by  1 / 1,0  multiplier                                      *
*                                                                                                                              *
*     I/O Operation                                                                                                            *
*     Number of records read via DD * or DD DATA:      1                                                                       *
*     148......65 DMY.......0 180......40 DMY.......0 DMY.......0 DMY.......0                                                  *
*                                                                                                                              *
*                                       Charge for step (w/o SYSOUT):          0,08                                        *
*------------------------------------------------------------------------------------------------------------------------------*
IEF237I 180  ALLOCATED TO SYS00001
IEF285I   SYS21330.T211721.RA000.PRIMFORH.R0000001     KEPT          *--------0
IEF285I   VOL SER NOS= WORK02.
```

```
IEF285I   SYS21330.T211720.RA000.PRIMFORH.LOADSET      DELETED
IEF285I   VOL SER NOS= WORK02.
IEF375I  JOB /PRIMFORH/ START 21330.2117
IEF376I  JOB /PRIMFORH/ STOP  21330.2117 CPU    0MIN 00.09SEC SRB    0MIN 00.02SEC
```

```
         COMPILER OPTIONS - NAME=  MAIN,OPT=00,LINECNT=50,SIZE=0000K,
                            SOURCE,EBCDIC,NOLIST,NODECK,LOAD,MAP,NOEDIT,NOID,NOXREF
                     C /*-----------------------------------------------------------------*/
                     C /*   Sieve of Eratosthenes.                                        */
                     C /*-----------------------------------------------------------------*/
                     C
                     C /*-----------------------------------------------------------------*/
                     C /*   Formats for output.                                           */
                     C /*-----------------------------------------------------------------*/
   ISN 0002      3       FORMAT (' ')
   ISN 0003      4       FORMAT (  3X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7,
                        11X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X, I7, 1X
                        1, I7, 1X, I7, 1X, I7)
   ISN 0004      5       FORMAT (' Sieve of Eratosthenes generated using OS/360 FORTRAN H L
                        1evel 21.8')
   ISN 0005      6       FORMAT (' Upper limit of test range = ', I12)
   ISN 0006      7       FORMAT (' Number of primes in range = ', I12)
   ISN 0007      8       FORMAT (I7)
   ISN 0008      993     FORMAT (' DEBUG: marking as non prime: ', I4)
   ISN 0009      994     FORMAT (' DEBUG: starting p=', I4, '    j=', I4)
                     C
                     C /*-----------------------------------------------------------------*/
                     C /*   Define array of flags, one for each integer in the range      */
                     C /*   we will test.  If the flag is on, the corresponding           */
                     C /*   number is prime.  If it's off, the number is not prime.       */
                     C /*   We will initialize all the flags to on (assuming every        */
                     C /*   number is prime) and turn them off as we determine the        */
                     C /*   corresponding number is not prime.                            */
                     C /*-----------------------------------------------------------------*/
   ISN 0010              LOGICAL*1 FLAGS(5000002)
                     C
                     C /*-----------------------------------------------------------------*/
                     C
                     C /*-----------------------------------------------------------------*/
                     C /*   The PRIME array will hold all the prime numbers we have       */
                     C /*   identified, and CPRIME will contain the number of primes      */
                     C /*   we've found.                                                  */
                     C /*-----------------------------------------------------------------*/
   ISN 0011              INTEGER*4 PRIME(350000)
   ISN 0012              INTEGER*4 CPRIME
                     C
                     C /*-----------------------------------------------------------------*/
                     C /*   J is a loop counter and work variable.                        */
                     C /*-----------------------------------------------------------------*/
   ISN 0013              INTEGER*4 J
                     C
                     C /*-----------------------------------------------------------------*/
                     C /*   K is the step amount for crossing out prime multiples         */
                     C /*-----------------------------------------------------------------*/
   ISN 0014              INTEGER*4 K
```

```
                C
                C /*-----------------------------------------------------------------*/
                C /*   P is the number that we've most recently determined           */
                C /*   definitely to be prime.                                       */
                C /*-----------------------------------------------------------------*/
   ISN 0015             INTEGER*4 P
                C
                C /*-----------------------------------------------------------------*/
                C /*   The DEBUG flag is set to TRUE if debugging messages are to    */
                C /*   be issued and FALSE otherwise.                                */
                C /*-----------------------------------------------------------------*/
   ISN 0016             LOGICAL*1 DEBUG
                C
                C /*-----------------------------------------------------------------*/
                C /*   LIMIT sets the upper bound of the range of numbers we         */
                C /*   will test.                                                    */
                C /*-----------------------------------------------------------------*/
   ISN 0017             INTEGER*4 LIMIT
                C
                C /*-----------------------------------------------------------------*/
                C /*   REPEAT is the number of times that the entire prime           */
                C /*   generation process is to be repeated, and is useful           */
                C /*   for benchmarking (otherwise it should be 1).                  */
                C /*-----------------------------------------------------------------*/
   ISN 0018             INTEGER*4 REPEAT
                C
                C /*-----------------------------------------------------------------*/
                C /*   Initialize LIMIT, DEBUG and REPEAT.                           */
                C /*-----------------------------------------------------------------*/
   ISN 0019             READ (5, 8) LIMIT
                C       DEBUG = .TRUE.
   ISN 0020             DEBUG = .FALSE.
   ISN 0021             REPEAT = 1
                C
                C *---------------------------------------------------------------*
                C *    THIS IS THE TOP OF THE LOOP FOR BENCHMARK TESTING.
                C *---------------------------------------------------------------*
   ISN 0022     100     CONTINUE
   ISN 0023             REPEAT = REPEAT - 1
   ISN 0024             IF (REPEAT .LT. 0) GO TO 999
                C
                C /*-----------------------------------------------------------------*/
                C /*   Initialize all flags to on.  We optimistically assume         */
                C /*   all numbers are prime, and will subsequently turn flags       */
                C /*   off as reality sets in.                                       */
                C /*-----------------------------------------------------------------*/
   ISN 0026             DO 200 J = 3, LIMIT, 2
   ISN 0027               FLAGS(J) = .TRUE.
```

```
ISN 0028     200   CONTINUE
                   C
                   C /*-----------------------------------------------------------------*/
                   C /*   The first prime number is 3, the 2 is handled manually        */
                   C /*-----------------------------------------------------------------*/
ISN 0029                 P = 3
                   C
                   C  /*-----------------------------------------------------------------*/
                   C  /*   Start of the main loop.  P is the prime number we're          */
                   C  /*   currently working on.  If P*P is greater than the limit       */
                   C  /*   value, we're done (all the numbers between P and the limit    */
                   C  /*   inclusive have already been marked appropriately).  Any       */
                   C  /*   non-prime less than P*P has also already been marked          */
                   C  /*   appropriately, so we will start this pass marking with        */
                   C  /*   P*P (which we will call J).                                   */
                   C  /*-----------------------------------------------------------------*/
ISN 0030     300   CONTINUE
ISN 0031           J = P * P
ISN 0032           K = 2 * P
ISN 0033           IF (J .GE. LIMIT) GO TO 700
ISN 0035             IF (.NOT. DEBUG) GO TO 400
ISN 0037               WRITE (6, 994) P, J
                   C
                   C /*-----------------------------------------------------------------*/
                   C /*   By definition, all multiples of prime number P are not        */
                   C /*   prime.  Turn off the flags for the multiples of P to          */
                   C /*   mark them as non-prime. Note: Even numbers are skipped.       */
                   C /*-----------------------------------------------------------------*/
ISN 0038     400   CONTINUE
ISN 0039             IF (J .GT. LIMIT) GO TO 500
ISN 0041             IF (.NOT. DEBUG) GO TO 420
ISN 0043               WRITE (6, 993) J
ISN 0044     420     FLAGS(J) = .FALSE.
ISN 0045             J = J + K
ISN 0046           GO TO 400
                   C
                   C /*-----------------------------------------------------------------*/
                   C /*   Done marking all multiples of J as not prime.  Find the       */
                   C /*   next prime number after J, set it to P and loop back to       */
                   C /*   process it. Note: Even numbers are skipped.                   */
                   C /*-----------------------------------------------------------------*/
ISN 0047     500   CONTINUE
ISN 0048             P = P + 2
ISN 0049             IF (FLAGS(P)) GO TO 600
ISN 0051           GO TO 500
                   C
                   C /*-----------------------------------------------------------------*/
                   C /*   Bottom of the main loop.                                      */
                   C /*-----------------------------------------------------------------*/
```

```
                C /*-----------------------------------------------------------------*/
    ISN 0052    600   CONTINUE
    ISN 0053          GO TO 300
                C
                C /*-----------------------------------------------------------------*/
                C /*   Bottom of the benchmark loop.                                 */
                C /*-----------------------------------------------------------------*/
    ISN 0054    700   CONTINUE
    ISN 0055          GO TO 100
    ISN 0056    999   CONTINUE
                C
                C /*-----------------------------------------------------------------*/
                C /*   Set the prime numbers we have found in the PRIME array.       */
                C /*-----------------------------------------------------------------*/
    ISN 0057          CPRIME = 1
    ISN 0058          PRIME(CPRIME) = 2
    ISN 0059          DO 800 J = 3, LIMIT, 2
    ISN 0060            IF (.NOT. FLAGS(J)) GO TO 800
    ISN 0062              CPRIME = CPRIME + 1
    ISN 0063              PRIME(CPRIME) = J
    ISN 0064    800   CONTINUE
                C
                C /*-----------------------------------------------------------------*/
                C /*   Display the results.                                          */
                C /*-----------------------------------------------------------------*/
    ISN 0065          WRITE (6, 3)
    ISN 0066          WRITE (6, 5)
    ISN 0067          WRITE (6, 6) LIMIT
    ISN 0068          WRITE (6, 7) CPRIME
    ISN 0069          WRITE (6, 3)
    ISN 0070          WRITE (6, 4) (PRIME(J), J = 1, CPRIME)
    ISN 0071          WRITE (6, 3)
                C
                C /*-----------------------------------------------------------------*/
                C /*   End of program.                                               */
                C /*-----------------------------------------------------------------*/
    ISN 0072          STOP
    ISN 0073          END
```

NAME	TAG	TYPE	ADD.	NAME	TAG	TYPE	ADD.	NAME	TAG	TYPE	ADD.	NAME	TAG	TYPE	ADD.
J	SF	I*4	0001B4	K	SF	I*4	0001B8	P	SF	I*4	0001BC	DEBUG	S	L*1	0001B0
FLAGS	S	L*1	0001CC	LIMIT	SF	I*4	0001C0	PRIME	SF	I*4	4C4D14	CPRIME	SF	I*4	0001C4
IBCOM#	F	XF	R*4	000000	REPEAT	SF	I*4	0001C8							

```
100  61AA6C          200  61AA92          300  61AAB0          400  61AB0C
420  61AB48          500  61AB66          600  61AB8C          700  61AB92
999  61AB98          800  61ABEC
```

```
*OPTIONS IN EFFECT*      NAME=  MAIN,OPT=00,LINECNT=50,SIZE=0000K,
*OPTIONS IN EFFECT*      SOURCE,EBCDIC,NOLIST,NODECK,LOAD,MAP,NOEDIT,NOID,NOXREF

*STATISTICS*     SOURCE STATEMENTS =      72 ,PROGRAM SIZE =  6401286
*STATISTICS*  NO  DIAGNOSTICS GENERATED
 ****** END OF COMPILATION ******                                  8917K BYTES OF CORE NOT USED
```

VS LOADER

OPTIONS USED - PRINT,NOMAP,NOLET,CALL,RES,NOTERM,SIZE=9000000,NAME=**GO

  TOTAL LENGTH   61FFD0
  ENTRY ADDRESS   ACBD0

1412THE

```
Sieve of Eratosthenes generated using OS/360 FORTRAN H Level 21.8
Upper limit of test range =         2000
Number of primes in range =          303

       2     3     5     7    11    13    17    19    23    29    31    37    41    43    47    53
      67    71    73    79    83    89    97   101   103   107   109   113   127   131   137   139
     157   163   167   173   179   181   191   193   197   199   211   223   227   229   233   239
     257   263   269   271   277   281   283   293   307   311   313   317   331   337   347   349
     367   373   379   383   389   397   401   409   419   421   431   433   439   443   449   457
     467   479   487   491   499   503   509   521   523   541   547   557   563   569   571   577
     599   601   607   613   617   619   631   641   643   647   653   659   661   673   677   683
     709   719   727   733   739   743   751   757   761   769   773   787   797   809   811   821
     829   839   853   857   859   863   877   881   883   887   907   911   919   929   937   941
     967   971   977   983   991   997  1009  1013  1019  1021  1031  1033  1039  1049  1051  1061
    1087  1091  1093  1097  1103  1109  1117  1123  1129  1151  1153  1163  1171  1181  1187  1193
    1217  1223  1229  1231  1237  1249  1259  1277  1279  1283  1289  1291  1297  1301  1303  1307
    1327  1361  1367  1373  1381  1399  1409  1423  1427  1429  1433  1439  1447  1451  1453  1459
    1483  1487  1489  1493  1499  1511  1523  1531  1543  1549  1553  1559  1567  1571  1579  1583
    1607  1609  1613  1619  1621  1627  1637  1657  1663  1667  1669  1693  1697  1699  1709  1721
    1741  1747  1753  1759  1777  1783  1787  1789  1801  1811  1823  1831  1847  1861  1867  1871
    1879  1889  1901  1907  1913  1931  1933  1949  1951  1973  1979  1987  1993  1997  1999


              PPPPPPPPPP   RRRRRRRRRR    IIIIIIIII   MM        MM  FFFFFFFFFF  OOOOOOOOOO  RRRRRRRRR   HH       HH
              PPPPPPPPPPP  RRRRRRRRRR   IIIIIIIII   MMM      MMM  FFFFFFFFFF  OOOOOOOOOO  RRRRRRRRRR  HH       HH
            PP        PP  RR       RR     II       MMMM    MMMM  FF          OO      OO  RR       RR  HH       HH
           PP         PP  RR       RR     II       MM MM  MM MM  FF          OO      OO  RR       RR  HH       HH
          PP          PP  RR       RR     II       MM  MMMM  MM  FF          OO      OO  RR       RR  HH       HH
         PPPPPPPPPPP  RRRRRRRRRR       II       MM   MM   MM  FFFFFFF       OO      OO  RRRRRRRRRR  HHHHHHHHHHH
        PPPPPPPPPP   RRRRRRRRR        II       MM        MM  FFFFFFF       OO      OO  RRRRRRRRR   HHHHHHHHHHH
       PP            RR    RR        II       MM        MM  FF          OO      OO  RR    RR      HH       HH
      PP             RR     RR       II       MM        MM  FF          OO      OO  RR     RR     HH       HH
     PP              RR      RR      II       MM        MM  FF          OO      OO  RR      RR    HH       HH
    PP               RR       RR  IIIIIIIII  MM        MM  FF          OOOOOOOOOO  RR       RR   HH       HH
   PP                RR        RR IIIIIIIII  MM        MM  FF          OOOOOOOOOO  RR        RR  HH       HH



                    JJJJJJJJJJ      11       3333333333                              AAAAAAAAAA
                    JJJJJJJJJJ     111      333333333333                            AAAAAAAAAAAA
                        JJ        1111      33       33                             AA        AA
                        JJ          11                33                             AA        AA
                        JJ          11                33                             AA        AA
                        JJ          11              3333                             AAAAAAAAAAAA
                        JJ          11              3333                             AAAAAAAAAAAA
                        JJ          11                33                             AA        AA
                  JJ    JJ          11                33                             AA        AA
                  JJ    JJ          11      33        33                             AA        AA
                  JJJJJJJJ      1111111111  333333333333                            AA        AA
                   JJJJJJ       1111111111   3333333333                             AA        AA

****A   END   JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13   END   A****
****A   END   JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13   END   A****
****A   END   JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13   END   A****
****A   END   JOB   13  PRIMFORH  Eratosthenes Sieve    ROOM        9.17.21 PM 26 NOV 21  PRINTER1  SYS TK4-  JOB   13   END   A****
```